Apresentação MESA 7

FortiLayer Security

Segurança cibernética

Overview

1. Conceito de Firewall no Contexto Industrial;
2. Diferença entre Firewall Corporativo e Industrial;
3. Proteção da Infraestrutura TI e OT;
4. Monitoramento
5. Topologia;
6. Cenário da Rede Industrial (Simulação):

Firewall no Contexto Industrial

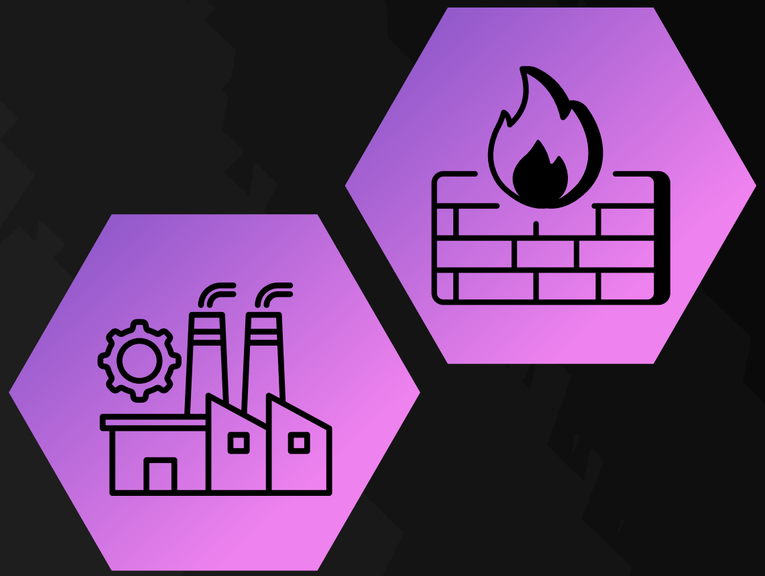
No ambiente industrial, o firewall é essencial para garantir que apenas comunicações autorizadas cheguem aos sistemas de produção.

Ele entende protocolos industriais e precisa funcionar em ambientes mais críticos e robustos.

Protege máquinas e sistemas de controle industrial, como CLPs, sensores e sistemas de supervisão

Precisa garantir comunicação segura e contínua entre os equipamentos da produção

E evitar falhas ou ataques que possam parar máquinas ou interromper a produção.



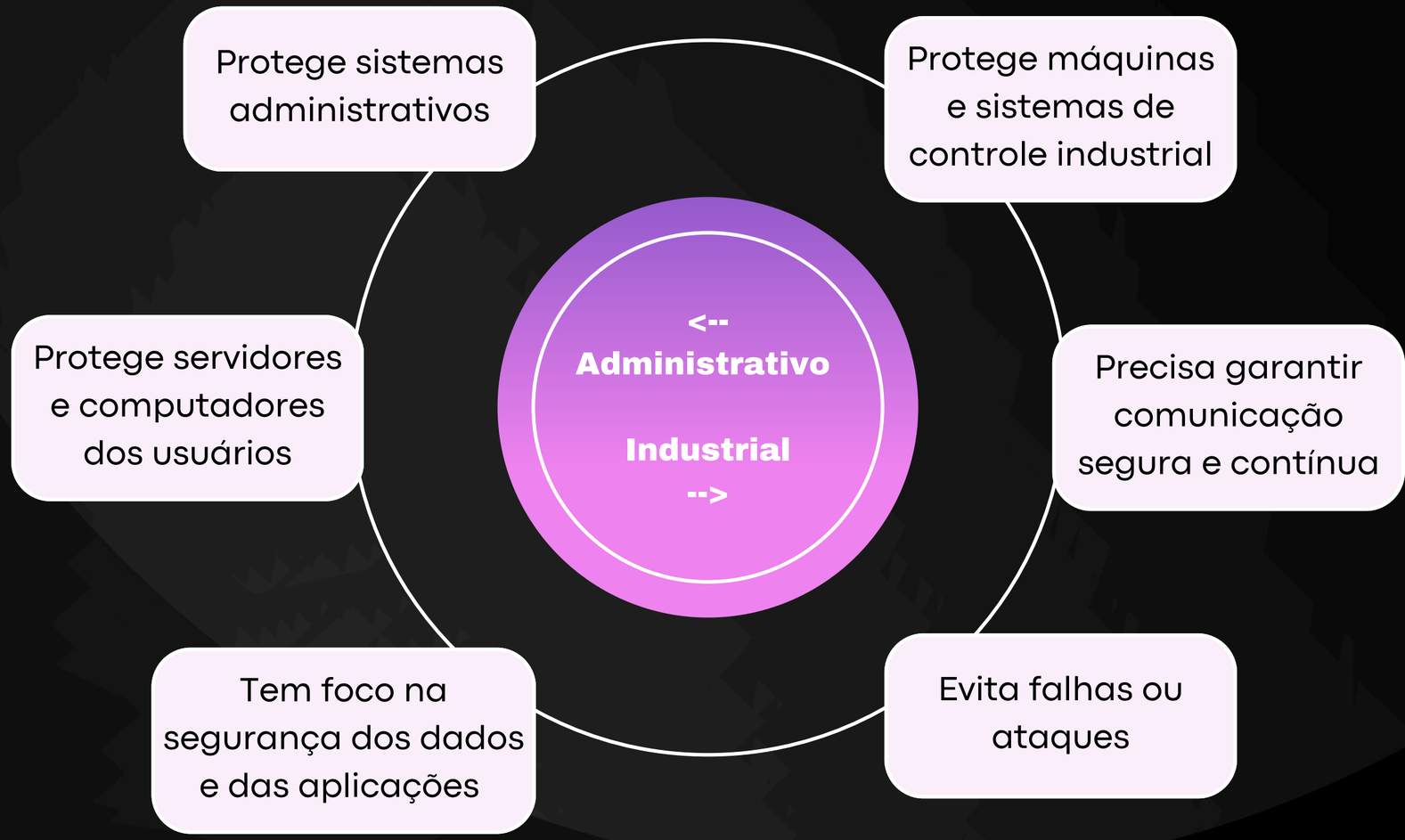
Firewall Corporativo

01 impede que pessoas não autorizadas acessem sistemas internos da empresa, como financeiro, RH ou gestão.

02 bloqueia ataques da internet que poderiam atingir os computadores dos funcionários ou os servidores da empresa.

03 ajuda a garantir que informações importantes e programas da empresa não sejam roubados, alterados ou acessados por quem não deve.

DIFERENÇAS



Proteção da Infraestrutura TI e OT

Infraestrutura de TI (Tecnologia da Informação):
Sistemas responsáveis pela gestão administrativa da empresa

Exemplos

- ERP (gestão empresarial)
- Servidores de arquivos
- Computadores administrativos

Função do firewall

- Controlar acessos entre redes
- Impedir comunicação não autorizada
- Proteger dados corporativos

Proteção da Infraestrutura TI e OT

A OT (Tecnologia Operacional): Envolve todos os sistemas que monitoram ou controlam processos físicos da indústria.

Exemplos

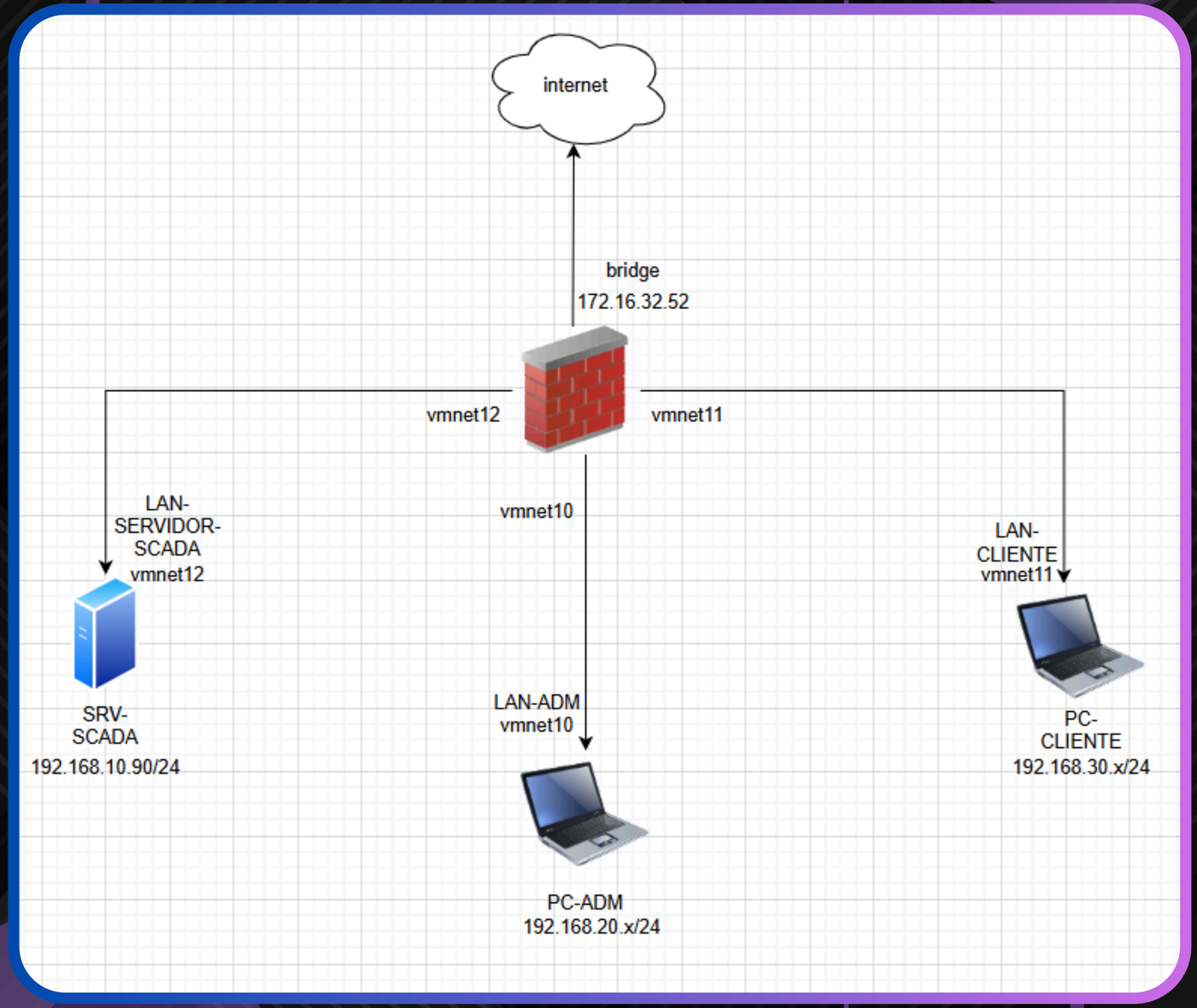
- Controladores industriais PLC
- Sistemas de supervisão SCADA
- Robôs industriais
- Sensores e equipamentos da linha de produção

Função do Firewall

- Impedir acesso direto da rede corporativa
- Permitir somente acessos autorizados
- Reduzir riscos de ataques à produção



TOPOLOGIA


internet
bridge
172.16.32.52
vmnet12
vmnet11
vmnet10
LAN-
SERVIDOR-
SCADA
vmnet12
SRV-
SCADA
192.168.10.90/24
LAN-ADM
vmnet10
PC-ADM
192.168.20.x/24
LAN-
CLIENTE
vmnet11
PC-
CLIENTE
192.168.30.x/24

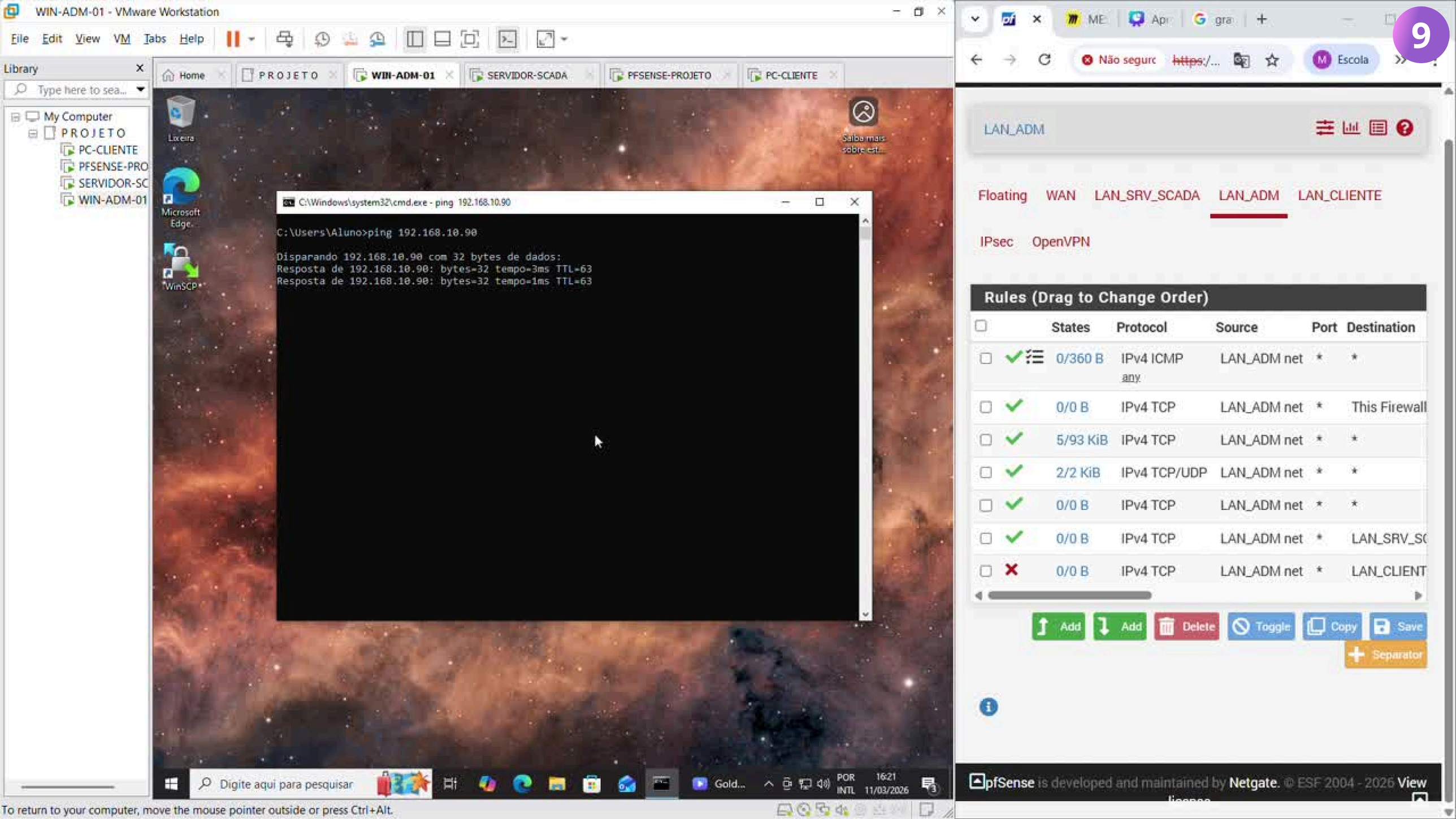
Regras LAN_ADM

Floating | WAN | LAN_SRV_SCADA | LAN_ADM | LAN_CLIENTE | IPsec | OpenVPN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/240 B	IPv4 ICMP any	LAN_ADM net	*	LAN_SRV_SCADA net	*	*	none		LAN TEM ACESSO AO ICMP	
✓	0/224 KiB	IPv4 TCP	LAN_ADM net	*	This Firewall	*	*	none		ADM ACESSA PFSENSE	
✓	0/28.38 MiB	IPv4 TCP	LAN_ADM net	*	*	443 (HTTPS)	*	none		ADM ACESSA HTTPS	
✓	0/43 KiB	IPv4 TCP/UDP	LAN_ADM net	*	*	53 (DNS)	*	none		ADM ACESSA DNS	
✓	0/2.43 MiB	IPv4 TCP	LAN_ADM net	*	*	80 (HTTP)	*	none		ADM ACESSA HTTP	
✓	0/4 KiB	IPv4 TCP	LAN_ADM net	*	LAN_SRV_SCADA net	22 (SSH)	*	none		SSH ADM ACESSA SCADA SSH	
✗	0/0 B	IPv4 TCP	LAN_ADM net	*	LAN_CLIENTE net	22 (SSH)	*	none		LAN ADM NAO ACESSA SSH LAN CLIENTE	

Add | Add | Delete | Toggle | Copy | Save | Separator


WIN-ADM-01 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
PROJETO
PC-CLIENTE
PFSENSE-PRO
SERVIDOR-SC
WIN-ADM-01
Home
PROJETO
WIN-ADM-01
SERVIDOR-SCADA
PFSENSE-PROJETO
PC-CLIENTE
Lixeira
Microsoft Edge
WinSCP
Saiba mais sobre est...
C:\Windows\system32\cmd.exe - ping 192.168.10.90
C:\Users\Aluno>ping 192.168.10.90
Disparando 192.168.10.90 com 32 bytes de dados:
Resposta de 192.168.10.90: bytes=32 tempo=3ms TTL=63
Resposta de 192.168.10.90: bytes=32 tempo=1ms TTL=63
Digite aqui para pesquisar
POR INTL
16:21
11/03/2026
To return to your computer, move the mouse pointer outside or press Ctrl+Alt.
Não seguro
Escola
LAN_ADM
Floating
WAN
LAN_SRV_SCADA
LAN_ADM
LAN_CLIENTE
IPsec
OpenVPN
Rules (Drag to Change Order)
States
Protocol
Source
Port
Destination
0/360 B
IPv4 ICMP any
LAN_ADM net
*
*
0/0 B
IPv4 TCP
LAN_ADM net
*
This Firewall
5/93 KiB
IPv4 TCP
LAN_ADM net
*
*
2/2 KiB
IPv4 TCP/UDP
LAN_ADM net
*
*
0/0 B
IPv4 TCP
LAN_ADM net
*
*
0/0 B
IPv4 TCP
LAN_ADM net
*
LAN_SRV_SC
0/0 B
IPv4 TCP
LAN_ADM net
*
LAN_CLIENT
Add
Add
Delete
Toggle
Copy
Save
Separator
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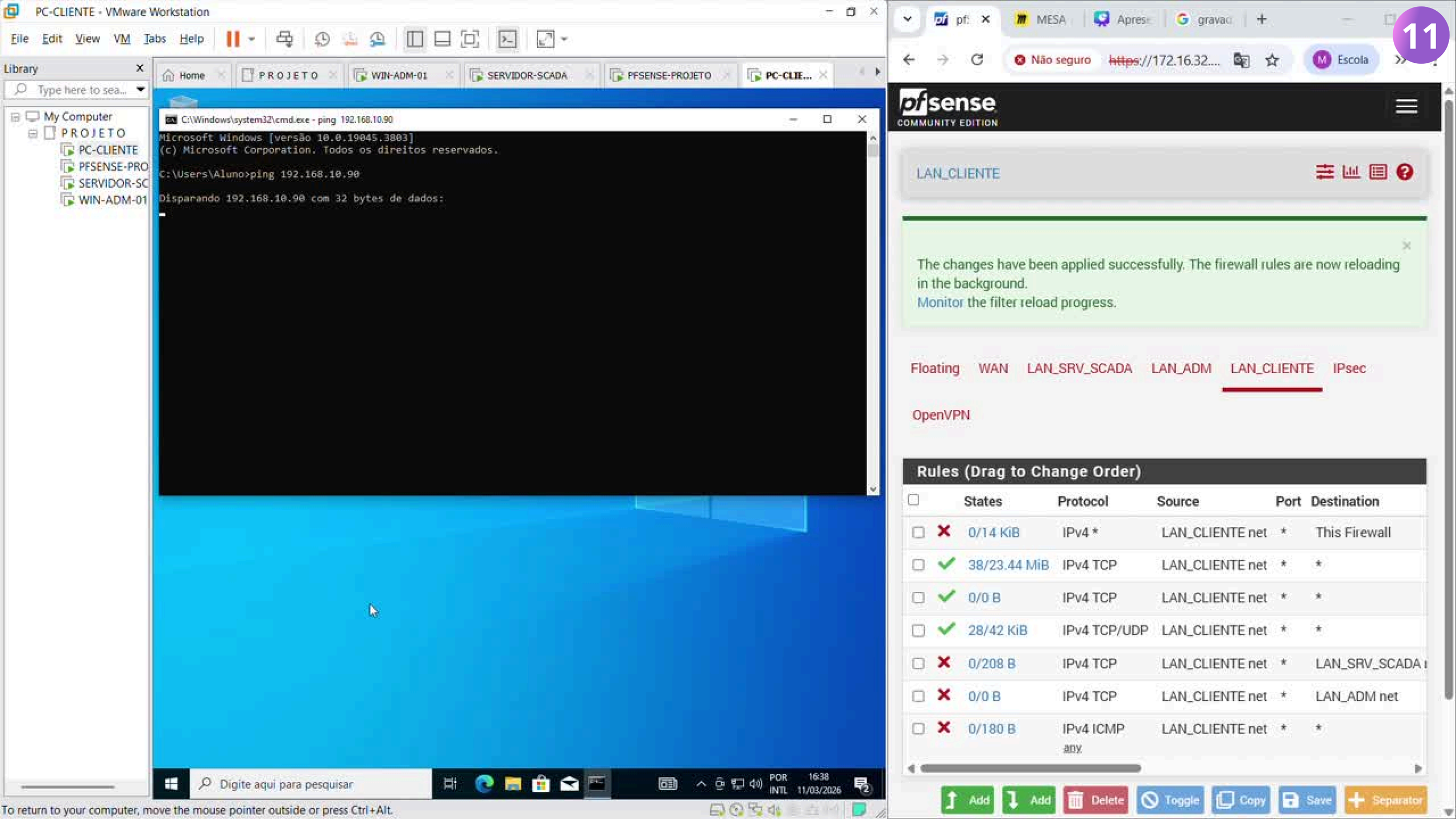
Regras LAN_Cliente

Floating | WAN | LAN_SRV_SCADA | LAN_ADM | LAN_CLIENTE | IPsec | OpenVPN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 ICMP any	LAN_CLIENTE net	*	*	*	*	none		CLIENTE ACESSA E PINGA ICMP	
☐	✓	0/137 KiB	IPv4 TCP	LAN_CLIENTE net	*	*	443 (HTTPS)	*	none		CLIENTE ACESSA HTTPS	
☐	✓	0/817 B	IPv4 TCP	LAN_CLIENTE net	*	*	80 (HTTP)	*	none		CLIENTE ACESSA HTTP	
☐	✓	0/8 KiB	IPv4 TCP/UDP	LAN_CLIENTE net	*	*	53 (DNS)	*	none		CLIENTE ACESSA DNS	
☐	✗	0/0 B	IPv4 TCP	LAN_CLIENTE net	*	LAN_SRV_SCADA net	22 (SSH)	*	none		CLIENTE NAO ACESSA SSH SCADA	
☐	✗	0/0 B	IPv4 TCP	LAN_CLIENTE net	*	LAN_ADM net	22 (SSH)	*	none			
☐	✗	0/0 B	IPv4 TCP	LAN_CLIENTE net	*	LAN_ADM net	22 (SSH)	*	none		CLIENTE NAO ACESSA SSH ADM	

Add | Add | Delete | Toggle | Copy | Save | Separator


PC-CLIENTE - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
PROJETO
PC-CLIENTE
PFSENSE-PRO
SERVIDOR-SC
WIN-ADM-01
Home
PROJETO
WIN-ADM-01
SERVIDOR-SCADA
PFSENSE-PROJETO
PC-CLIE...
C:\Windows\system32\cmd.exe - ping 192.168.10.90
Microsoft Windows [versão 10.0.19045.3803]
(c) Microsoft Corporation. Todos os direitos reservados.
C:\Users\Aluno>ping 192.168.10.90
Disparando 192.168.10.90 com 32 bytes de dados:
Digite aqui para pesquisar
POR INTL
16:38
11/03/2026
To return to your computer, move the mouse pointer outside or press Ctrl+Alt.
pf:
MESA
Apres
gravac
11
Não seguro
https://172.16.32....
Escola
pfsense
COMMUNITY EDITION
LAN_CLIENTE
The changes have been applied successfully. The firewall rules are now reloading in the background.
Monitor the filter reload progress.
Floating
WAN
LAN_SRV_SCADA
LAN_ADM
LAN_CLIENTE
IPsec
OpenVPN
Rules (Drag to Change Order)
States
Protocol
Source
Port
Destination
0/14 KiB
IPv4 *
LAN_CLIENTE net
*
This Firewall
38/23.44 MiB
IPv4 TCP
LAN_CLIENTE net
*
*
0/0 B
IPv4 TCP
LAN_CLIENTE net
*
*
28/42 KiB
IPv4 TCP/UDP
LAN_CLIENTE net
*
*
0/208 B
IPv4 TCP
LAN_CLIENTE net
*
LAN_SRV_SCADA
0/0 B
IPv4 TCP
LAN_CLIENTE net
*
LAN_ADM net
0/180 B
IPv4 ICMP any
LAN_CLIENTE net
*
*
Add
Add
Delete
Toggle
Copy
Save
Separator

Regras LAN_SCADA

Floating | WAN | LAN_SRV_SCADA | LAN_ADM | LAN_CLIENTE | IPsec | OpenVPN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/0 B	*	*	*	LAN_SRV_SCADA Address	443 80	*	*		Anti-Lockout Rule	⚙
☐ ✗	0/0 B	IPv4 TCP	LAN_SRV_SCADA net	*	LAN_CLIENTE net	22 (SSH)	*	none		SCADA NAO ACESSA SSH CLIENTE	⚓ ✎ ⧉ ⊘ 🗑
☐ ✓	0/3 KiB	IPv4 TCP/UDP	LAN_SRV_SCADA net	*	*	53 (DNS)	*	none		SCADA ACESSA DNS	⚓ ✎ ⧉ ⊘ 🗑✗
☐ ✓	0/8 KiB	IPv4 TCP	LAN_SRV_SCADA net	*	*	443 (HTTPS)	*	none		SRVSCADA ACESSA HTTPS	⚓ ✎ ⧉ ⊘ 🗑✗
☐ ✓	0/0 B	IPv4 TCP	LAN_SRV_SCADA net	*	*	80 (HTTP)	*	none		SRV SCADA ACESSA HTTP	⚓ ✎ ⧉ ⊘ 🗑✗
☐ ✗	0/0 B	IPv4 TCP	LAN_SRV_SCADA net	*	LAN_ADM net	22 (SSH)	*	none		SRV SCADA NAO ACESSA SSH LAN ADM	⚓ ✎ ⧉ ⊘ 🗑

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SERVIDOR-SCADA - VMware Workstation

File Edit View VM Tabs Help

Library

Type here to sea...

My Computer
- PROJETO
  - PC-CLIENTE
  - PFSENSE-PRO
  - SERVIDOR-SC
  - WIN-ADM-01

Home | PROJETO | WIN-ADM-01 | SERVIDOR-SCADA | PFS...

```
root@debian:~# ip -c a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qle
    link/ether 00:0c:29:b9:1b:78 brd ff:ff:ff:ff:ff:ff
    altname enp2s2
    inet 192.168.10.94/24 brd 192.168.10.255 scope global dynamic noprefixroute ens34
       valid_lft 3932sec preferred_lft 3932sec
    inet6 fe80::20c:29ff:fe[illegible]/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
root@debian:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=6.67 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=4.32 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=4.39 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=4.03 ms
^C
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 4.030/4.853/6.673/1.059 ms
root@debian:~# ping 192.168.20.100
PING 192.168.20.100 (192.168.20.100) 56(84) bytes of data.
^C
--- 192.168.20.100 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2050ms

root@debian:~# ping 192.168.30.100
PING 192.168.30.100 (192.168.30.100) 56(84) bytes of data.
^C
--- 192.168.30.100 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3074ms

root@debian:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=3.88 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=3.91 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=4.26 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=4.13 ms
^C
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 3.879/4.044/4.262/0.158 ms
root@debian:~# ping
```

To return to your computer, press Ctrl+Alt.

pfSense.h | MESA 7 - | Apresent | gravador

Não seguro https://172.16.32.52/firewall_r...

Escola

13

The changes have been applied successfully. The firewall rules are now reloading in the background.
Monitor the filter reload progress.

Floating | WAN | LAN_SRV_SCADA | LAN_ADM | LAN_CLIENTE | IPsec | OpenVPN

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/0 B	*	*	*	LAN_SRV_SCADA Address	443 80	*	*		Anti-Lockout Rule	
✗	0/0 B	IPv4 ICMP any	LAN_SRV_SCADA net	*	LAN_ADM net	*	*	none		BLOQUEAR PING DO SCADA NO ADM	
✗	0/3 KiB	IPv4 ICMP any	LAN_SRV_SCADA net	*	LAN_CLIENTE net	*	*	none		BLOQUEAR PING DO SCADA NO CLIENTE	
✗	0/0 B	IPv4 TCP	LAN_SRV_SCADA net	*	LAN_CLIENTE net	22 (SSH)	*	none		SCADA NAO ACESSA SSH CLIENTE	
✗	0/0 B	IPv4 TCP	LAN_SRV_SCADA net	*	LAN_ADM net	22 (SSH)	*	none		SRV SCADA NAO ACESSA SSH LAN ADM	
✓	0/504 B	IPv4 ICMP any	LAN_SRV_SCADA net	*	*	*	*	none		PING NA INTERNET	
✓	0/3 KiB	IPv4 TCP/UDP	LAN_SRV_SCADA net	*	*	53 (DNS)	*	none		SCADA ACESSA DNS	
✓	0/15 KiB	IPv4 TCP	LAN_SRV_SCADA net	*	*	443 (HTTPS)	*	none		SRVSCADA ACESSA HTTPS	
✓	0/0 B	IPv4 TCP	LAN_SRV_SCADA net	*	*	80 (HTTP)	*	none		SRV SCADA ACESSA HTTP	

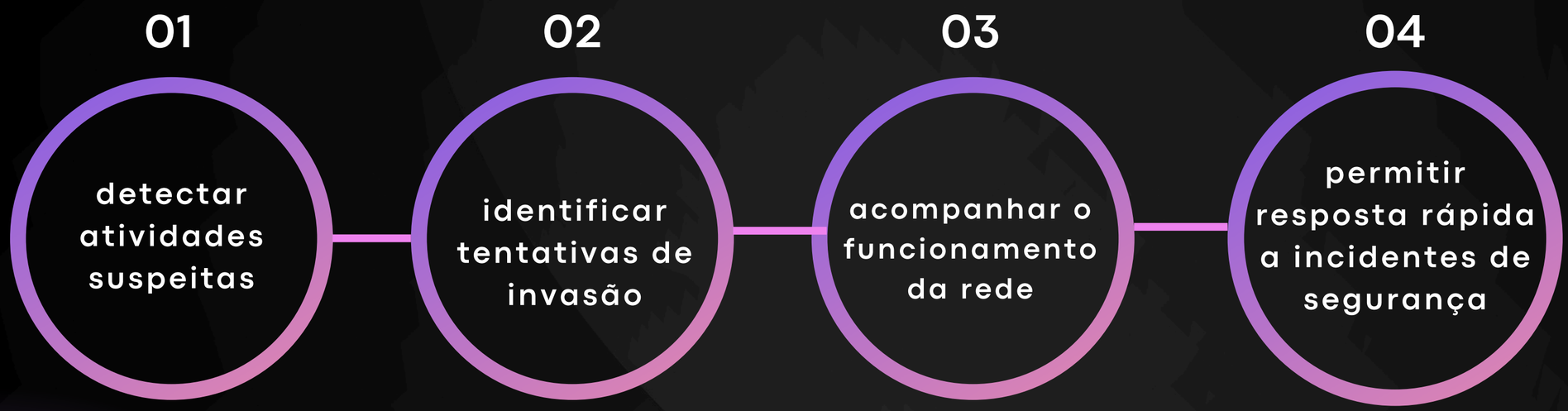
Add | Add | Delete | Toggle | Copy | Save | Separator

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SURICATA

Benefício do Monitoramento em Rede Industrial e Corporativa

O Suricata permite monitorar o tráfego da rede em tempo real, analisando os pacotes que passam pela infraestrutura de TI e OT para identificar atividades suspeitas ou maliciosas. Ele atua como um sistema de detecção e prevenção de intrusões (IDS/IPS), ajudando a proteger a rede contra ataques, malware e acessos não autorizados.



Em ambientes industriais, o monitoramento também ajuda a evitar falhas que possam interromper a produção.

All alert log files for selected interface will be downloaded

Clear

Clear the currently active Alerts log file

Save Settings

Save

Save auto-refresh and view settings

Refresh

Default is ON

250

Number of alerts to display. Default is 250

Alert Log View Filter

Last 250 Alert Entries. (Most recent entries are listed first)

Note: Alerts triggered by DROP rules that resulted in dropped (blocked) packets are shown with highlighted rows below.

Date	Action	Pri	Proto	Class	Src	SPort	Dst	DPort	GID
03/11/2026 19:39:34		2	TCP	Attempted Information Leak	172.16.32.94	48148	172.16.32.52	5810	1:2
03/11/2026 19:39:34		2	TCP	Potentially Bad Traffic	172.16.32.94	48150	172.16.32.52	1521	1:2
03/11/2026 19:39:34		2	TCP	Potentially Bad Traffic	172.16.32.94	48148	172.16.32.52	1521	1:2

Type here to sea...

My Computer
- FreeBSD version
- Windows 10 x64
- linux
- debian[

```
|_http-title: Did not follow redirect to https://172.16
443/tcp open  ssl/http nginx
|_http-title: pfSense - Login
| tls-alpn:
|   h2
|   http/1.1
|   http/1.0
|_  http/0.9
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject: commonName=pfSense-69af18d84794b/o
Signed Certificate
| Subject Alternative Name: DNS:pfSense-69af18d84794b
| Not valid before: 2026-03-09T19:00:40
|_Not valid after:  2027-04-11T19:00:40
1 service unrecognized despite returning data. If you k
llowing fingerprint at https://nmap.org/cgi-bin/submit.
SF-Port53-TCP:V=7.93%I=7%D=3/11%Time=69B1C4AA%P=x86_64-
SF:tatusRequestTCP,E,"\0\x0c\0\0\x90\x05\0\0\0\0\0\0\
MAC Address: 00:0C:29:94:54:C3 (VMware)
Warning: OSScan results may be unreliable because we co
t
Device type: general purpose
Running: FreeBSD 11.X
OS CPE: cpe:/o:freebsd:freebsd:11.2
OS details: FreeBSD 11.2-RELEASE
Network Distance: 1 hop

TRACEROUTE
HOP RTT     ADDRESS
1   1.71 ms 172.16.32.52

OS and Service detection performed. Please report any i
.
Nmap done: 1 IP address (1 host up) scanned in 37.17 se
root@debian:~# nmap -A 172.16.32.52
Starting Nmap 7.93 ( https://nmap.org ) at 2026-03-11 1
```

15

Firewall: Bloqueando ameaças na rede

OBRIGADA

por assistir nossa
apresentação

Mesa 7

Creditos:

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Matheus Maciel ZIravello

Pedro Henrique Brito

Julia Bueno de Andrade

Giulia Ferreira Silva